EMPLOYEE ONBOARDING & TASK HUB

A PROJECT REPORT

Submitted by

SHRIVARTHINI I

241401100

in partial fulfilment of the End Semester Examination for the course

CB23F35 –

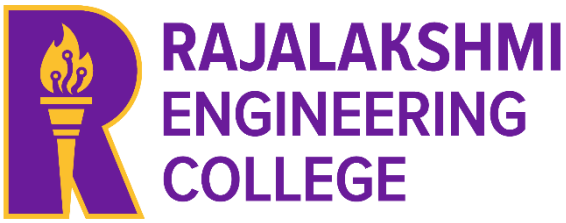
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND BUSINESS SYSTEM



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "EMPLOYEE ONBOARDING & TASK HUB" is the bonafide work of SHRIVARTHINI I [241401100], who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SHRIVARTHINI I (241401100), hereby declare that the project report titled "EMPLOYEE ONBOARDING & TASK HUB", submitted for the End Semester Examination of CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me. The application described in this report was developed using Zoho Creator based on the implementation details and testing observations recorded during the project.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Zoho Creator documentation and other external resources used in preparing this report are acknowledged in the references.

Place: Chennai
Date: [3/11/2026]

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SHRIVARTHINI I

ABSTRACT

Employee onboarding involves collecting employee details, defining onboarding activities, assigning tasks and monitoring completion. When these activities are tracked through separate spreadsheets, paper records and email conversations, HR personnel may face fragmented information, repeated data entry, delayed follow-up and limited visibility into onboarding progress.

The Employee Onboarding & Task Hub is a web-based application developed using Zoho Creator. It contains three forms—Employee_Details, Onboarding_Tasks and Task_Assignment—linked through lookup fields. Six list reports and an HR_Dashboard provide operational views. Three workflows automate welcome emails, completion-date population and due-date reminders. Deluge scripting is used to populate Completion_Date with the current date when an assignment is marked Completed.

Testing recorded successful employee creation, welcome-email delivery, employee-name insertion, task assignment, automatic completion-date population, due-date reminder delivery and dashboard display. The dashboard sample-data snapshot showed 26 employees, 13 onboarding tasks and 46 task assignments, including 30 completed tasks. These are sample record counts, not measured performance improvements. Production deployment, detailed role permissions and formal performance/security testing were not confirmed and are not claimed as completed.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Employee Onboarding; Task Management

PROJECT AT A GLANCE

Attribute	Details
Project Title	Employee Onboarding & Task Hub
Application Domain	Human Resource Operations / Employee Onboarding
Student Name & Register No.	Shrivarthini I – [241401100]
Zoho Creator Edition	CB23F35 – Low-Code Application Development using Zoho Creator
Application Name (Link Name)	Employee Onboarding & Task Hub
Published / Live URL	https://shrivarthinii2024csbs1163.zohocreatorportal.in
Demo Credentials (per role)	HR Administrator,Employee
Number of Forms / Subforms	3 Forms / 0 Subforms
Number of Reports (by type)	List: 6 | Kanban: 1 | Calendar: 1 | Pivot: 1 | Timeline: 1 | Total: 10
Number of Pages / Dashboards	1 Dashboard – HR_Dashboard
Number of Workflows	Form: 2 | Schedule: 1 | Approval: 0 | Blueprint: 0 | Batch: 0
Deluge Functions / Custom Actions	3 workflow automation scripts (welcome email, automatic completion date, due-date reminder).
Integrations	Zoho Creator email notification functionality; no external API integration confirmed
Roles & Profiles Defined	Roles: Employee, HR Administrator. Permission Profiles: Employee_Permissions, HR_Admin_Permissions
Environments Used	Zoho Creator development environment → Portal access configured. Separate staging/production environments not confirmed.
Date of Production Deployment	31/10/2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
CRUD	Create, Read, Update and Delete
DFD	Data Flow Diagram
ER	Entity–Relationship
HR	Human Resources
KPI	Key Performance Indicator
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UI	User Interface
URL	Uniform Resource Locator
Zoho	Zoho Corporation platform name
Deluge	Data Enriched Language for the Universal Grid Environment
API	Application Programming Interface
ID	Identifier

CHAPTER 1
INTRODUCTION

1.1 Background

Employee onboarding is the structured process through which a newly appointed employee is introduced to an organisation, its policies, work environment and initial responsibilities. The process commonly includes collecting employee details, identifying onboarding activities, assigning tasks, defining due dates and monitoring progress. HR personnel need accurate employee records and a clear view of outstanding activities to coordinate this process. This project models the process as a focused HR application; it does not represent a complete HR management system.

1.2 Problem Statement

Traditional onboarding may rely on separate spreadsheets, paper records and email exchanges. These disconnected methods can fragment employee information and make it difficult to coordinate assignments, deadlines and completion status. A centralised application is required to maintain employee and task records, link each assignment to the relevant employee and onboarding task, monitor progress and provide timely notifications.

- Employee information and onboarding task definitions may be stored separately, increasing repeated data entry.
- Task ownership, due dates and completion status may be difficult to monitor in one place.
- Manual follow-up can delay reminders and reduce visibility of incomplete onboarding activities.

1.3 Objectives

- To create structured forms for employee details, onboarding task definitions and task assignments.
- To automate welcome emails, completion-date population and due-date reminders using Zoho Creator workflows and Deluge.
- To provide list reports and a dashboard for monitoring employee, task and assignment information.

- To use Zoho Creator email functionality for employee notifications; no external API integration is claimed.
- To test the implemented forms, reports and workflow automations and identify deployment and governance items requiring further verification.

1.4 Scope of the Project

In scope: Employee_Details, Onboarding_Tasks and Task_Assignment forms; lookup relationships; six list reports; HR_Dashboard; welcome email, automatic completion-date and due-date reminder workflows. Out of scope or not verified: employee self-service portal, external HRMS/API integration, approval processes, advanced role-based permissions, mobile-specific configuration and production deployment controls.

1.5 Justification for a Low-Code Approach

Zoho Creator was selected because it provides visual form creation, lookup fields, reports, pages and workflow automation within one low-code environment. Deluge adds application-specific logic where configuration alone is insufficient. This allowed the project to implement a connected HR data model and selected email automations without developing a separate full-stack application.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The project addresses a common manual onboarding arrangement in which HR personnel maintain employee details, task lists and assignment progress through separate records or communication channels. A typical sequence involves collecting employee information, identifying required activities, allocating tasks, communicating due dates, following up on completion and preparing progress summaries.

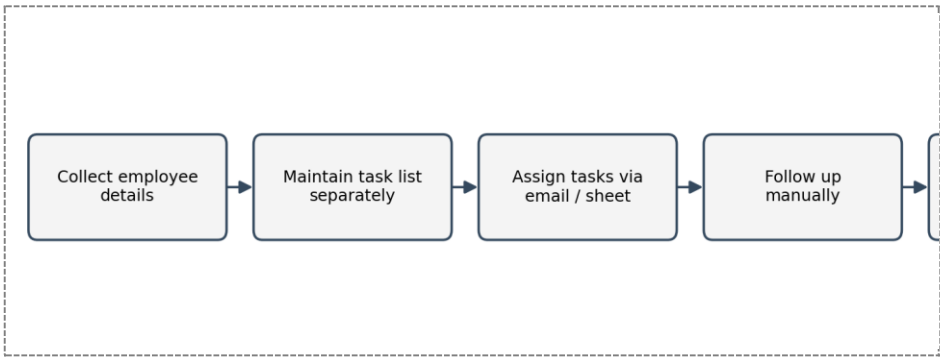


Figure 2.1: As-Is Employee Onboarding Process Flow

2.2 Limitations of the Existing System

- Employee and task information can be fragmented across separate files or communication channels.
- Assignment status and due dates may require manual consolidation and follow-up.
- There may be no single dashboard showing employee, task and completion summaries.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Requires separate UI, backend, database and deployment components.	Forms, reports, pages and workflows are configured in one platform.
Skills required	Programming, database, frontend and backend skills.	Visual configuration with Deluge for custom logic.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Data management	Database schema and CRUD interfaces must be built.	Creator forms and lookup fields provide data capture and relationships.
Automation	Requires custom backend jobs or notification services.	Creator workflows automate email and date update actions.
Reporting	Reports and dashboards must be developed separately.	List reports, charts and dashboard components are configured in Creator.
Maintenance	Codebase and infrastructure require separate maintenance.	Application configuration and Deluge scripts are maintained in Creator.
Project fit	Flexible but greater development overhead for this small workflow.	Suitable for a focused onboarding tracker with native automation.
Observed limitation	Greater control, but more components to build and integrate.	Platform-specific capabilities and configuration limits apply.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation	Understanding Forms	Creator forms capture and store application records.	Supports Employee_Details, Onboarding_Tasks and Task_Assignment.
2	Zoho Corporation	Understand Lookup Field	Lookup fields connect a record to another form record.	Supports the two lookup relationships in Task_Assignment.
3	Zoho Corporation	Understanding Pages	Pages combine application components for user-facing views.	Supports the HR_Dashboard page.
4	Zoho Corporation	Understanding Reports in a Page	Reports can be embedded in Creator pages.	Supports Pending Tasks and Completed Tasks dashboard panels.
5	Zoho Corporation	Accessing Form Fields using input	Deluge accesses and updates form fields through input.	Supports Completion_Date update script.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
6	Zoho Corporation	System Variables / Schedules	System date values and date-based schedules support automation.	Supports completion date and due-date reminder workflows.

2.5 Emerging Trends in Low-Code Development

The project demonstrates relevant low-code trends through visual application building, workflow automation and AI-assisted dashboard generation. Zoho Creator Zia AI was used to generate the HR_Dashboard. The project does not claim predictive analytics or autonomous HR decision-making. Low-code development can shorten configuration effort, while Deluge scripting provides control over application-specific workflow behaviour.

2.6 Proposed System

The proposed Employee Onboarding & Task Hub centralises employee records, reusable onboarding task definitions and employee-specific assignments. The Task_Assignment form uses two lookup fields to link each assignment to one employee and one onboarding task. Reports provide all-record and status-oriented views. The HR_Dashboard displays summary metrics, charts and embedded task reports. Workflows automate employee welcome emails, completion-date updates and due-date reminders.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
HR Administrator / Application User	Primary operational user represented by the project.	Maintain employee records, onboarding task definitions and task assignments; review reports and dashboard.
Employee	Notification recipient in the implemented workflows.	Receives welcome email and due-date reminder; employee login or self-service portal is not confirmed.
Project Developer	Person configuring the Creator application.	Create forms, lookup fields, reports, dashboard and workflow automations.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow employee details to be entered and stored in Employee_Details.	HR user	High
FR-02	The system shall allow onboarding task definitions to be stored in Onboarding_Tasks.	HR user	High
FR-03	The system shall allow task assignments to reference one employee and one onboarding task using lookup fields.	HR user	High
FR-04	The system shall store Assigned_Date, Due_Date, Completion_Date, Task_Status and Remarks for assignments.	HR user	High
FR-05	The system shall provide All_Employees, All_Onboarding_Tasks and All_Task_Assignments reports.	HR user	High
FR-06	The system shall provide Pending_Tasks, Completed_Tasks and Overdue_Tasks reports.	HR user	High; overdue filter to verify
FR-07	The system shall send a welcome email after a new Employee_Details record is submitted.	Employee / HR user	High

FR ID	Requirement	Role(s)	Priority
FR-08	The system shall populate Completion_Date with the current date when an edited assignment is marked Completed.	HR user	High
FR-09	The system shall send a due-date reminder for an assignment that is not Completed.	Employee / HR user	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Usability	Forms and reports should present employee and assignment information clearly.
NFR-02	Data integrity	Assignment lookups should reference existing employee and onboarding task records.
NFR-03	Maintainability	Form, report and workflow names should remain consistent and understandable.
NFR-04	Availability	The Creator application requires an internet connection and access to the configured Zoho account.
NFR-05	Notification reliability	Configured email workflows should deliver messages to the employee email address.
NFR-06	Security	User and record permissions should be reviewed and configured before operational deployment.
NFR-07	Performance	No measured response-time or load-test target was recorded; performance testing remains to be completed.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (exact subscription edition to be confirmed)
Scripting language	Deluge
Browser	Modern web browser with internet connectivity
Client device	Desktop or laptop capable of accessing Zoho Creator; exact specification not recorded
Network	Internet connection
Email	Valid employee email address for notification testing
Additional software	No separate database server or external API service is reported

3.5 Use Case Model

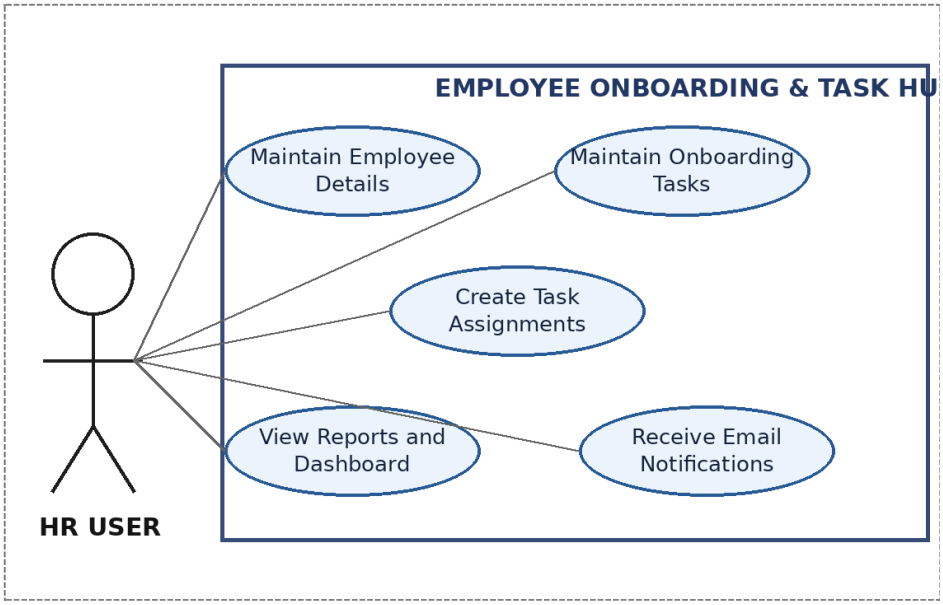


Figure 3.1: Use Case Diagram of Employee Onboarding & Task Hub

Table 3.5: Use Case Description – Create Employee and Assign Onboarding Task

Use Case ID	UC-01
Use Case Name	Create employee and assign onboarding task
Actor(s)	HR Administrator / Application User
Precondition	Employee and onboarding task records exist before assignment.
Main Flow	Open Task_Assignment; select an employee and task using the lookup fields; enter assignment dates, status and remarks; submit the record.
Alternative Flow	If the employee or task does not exist, create the required master record before assigning.
Postcondition	A Task_Assignment record is stored and linked to one employee and one onboarding task.
Related Workflows	Due_Date_Reminder and Auto_Set_Completion_Date

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

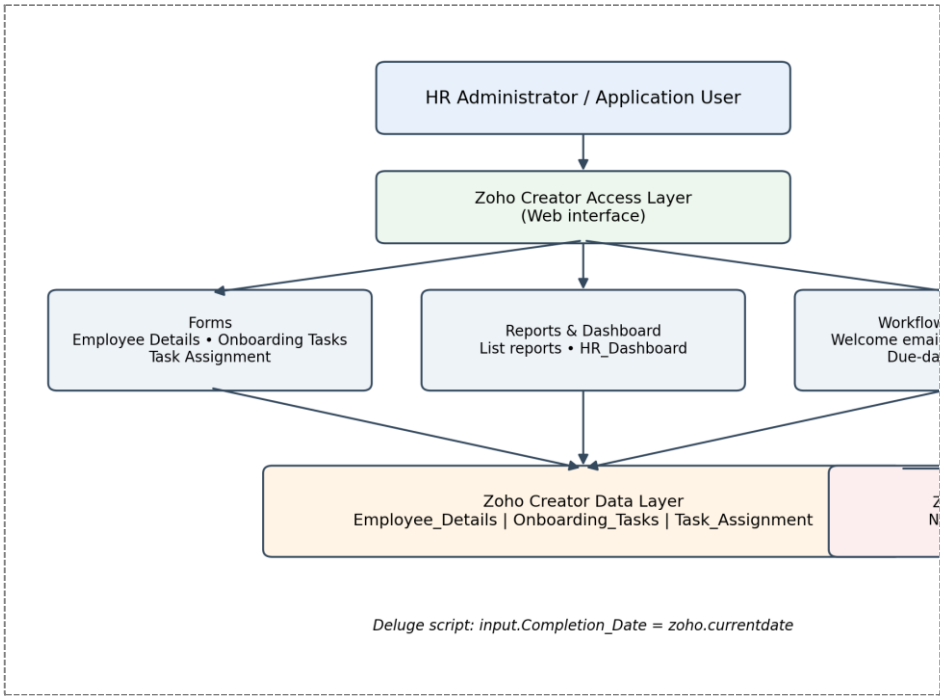


Figure 4.1: System Architecture of Employee Onboarding & Task Hub

The architecture is organised around the Zoho Creator application. HR/application users interact through the web interface. Forms provide data capture; reports and the HR_Dashboard provide data views; workflows execute notification and date-update actions. Employee_Details and Onboarding_Tasks act as master data forms, while Task_Assignment stores the employee-task relationship. Email notifications use Creator email functionality.

4.2 Data Model

The logical data model contains three forms. Employee_Details and Onboarding_Tasks store master data. Task_Assignment resolves the many-to-many association between employees and reusable onboarding tasks. Each Task_Assignment record references one employee and one task through two lookup fields. The exact link names and data types for fields other than the supplied names must be checked against the live Creator forms. Lookup fields connect a form record to a related record [2].

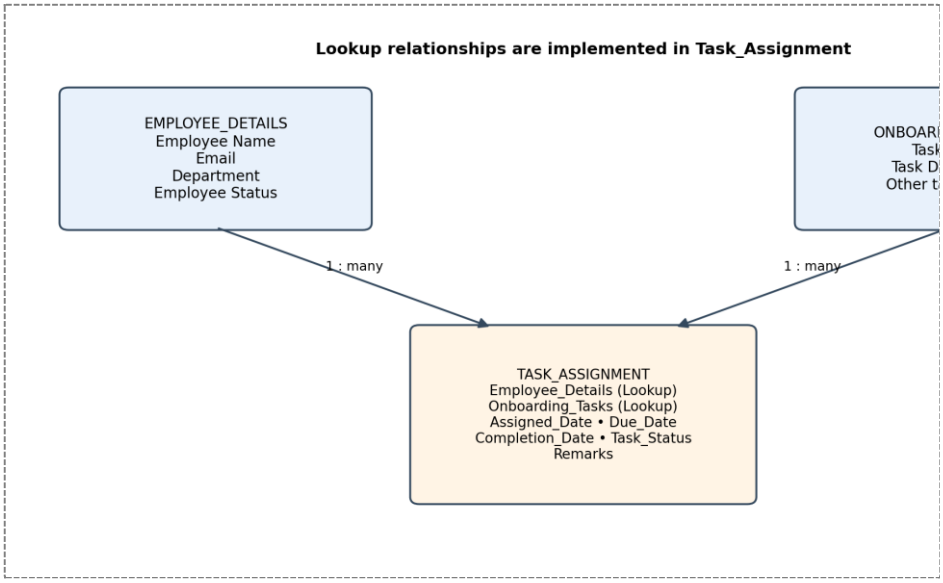


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Employee_Details	Employee Name; Email; Department; Employee Status	Task_Assignment	One-to-Many	Lookup in Task_Assignment
Onboarding_Tasks	Task Name; Task Description; additional task-related fields to verify	Task_Assignment	One-to-Many	Lookup in Task_Assignment
Task_Assignment	Employee_Details; Onboarding_Tasks; Assigned_Date; Due_Date; Completion_Date; Task_Status; Remarks	Employee_Details and Onboarding_Tasks	Many-to-One to each master form	Two lookup fields

4.3 Form Design

Table 4.2: Field Specification – Employee_Details, Onboarding_Tasks and Task_Assignment Forms

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
EMPLOYEE_DETAILS FORM					

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Employee Name	Employee Name	Name	yes	Required Field	Known field label; exact link name/type not supplied
Email	Email	Email	yes	Required field	Used as notification destination
Department	Department	Department	no	deafault	Used in department distribution chart
Employee Status	Employee Status	Employee Status	yes	validation	Exact options and field type to verify
ONBOARDING_TASKS FORM					
Task Name	Task Name	Name	yes	deafault	Identifies the onboarding activity
Task Description	Task Description	Single line	no	default	Additional task-related fields to verify
TASK_ASSIGNMENT FORM					
Employee_Details	Employee_Details	Lookup	no	Select existing employee record	Lookup to Employee_Details
Onboarding_Tasks	Onboarding_Tasks	Lookup	no	Select existing onboarding task	Lookup to Onboarding_Tasks
Assigned_Date	Assigned_Date	Date	yes	Required field	Date assigned
Due_Date	Due_Date	Date	yes	Required; cannot be earlier than Assigned_Date	Due date for reminder
Completion_Date	Completion_Date	Date	yes	Set by workflow when Completed	Current date assigned by Deluge
Task_Status	Task_Status	Drop down	no	Not Started / In Progress / Completed / On Hold	Used in reports and workflow conditions
Remarks	Remarks	Multi line	no	remarks	Assignment notes

Table 4.2 consolidates the known field labels for all three forms. Exact field link names, Creator field types, mandatory settings and validation rules were not supplied for several fields; these are marked for verification instead of being assumed. The lookup fields in Task_Assignment are confirmed as links to Employee_Details and Onboarding_Tasks.

4.4 Process and Workflow Design

Three automations are part of the implemented process. A new employee submission triggers a welcome email. Editing a task assignment to Completed triggers the completion-date update. A date-based schedule checks the due date and sends a reminder when the assignment is not Completed. Figure 4.3 illustrates the due-date reminder sequence; the other two workflows are documented in Table 5.3.

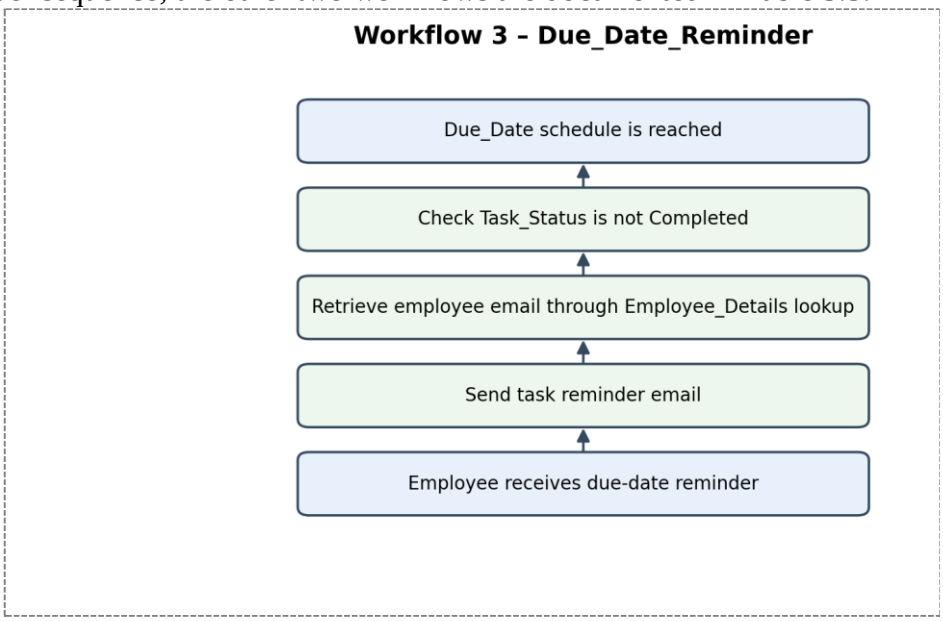


Figure 4.3: Due-Date Reminder Workflow Process

4.5 User Interface and Navigation Design

The application navigation is organised around employee records, onboarding task definitions, task assignments, the six list reports and HR_Dashboard. Figure 4.4 shows the logical navigation structure. The final menu labels should be compared with the live Creator application before submission.

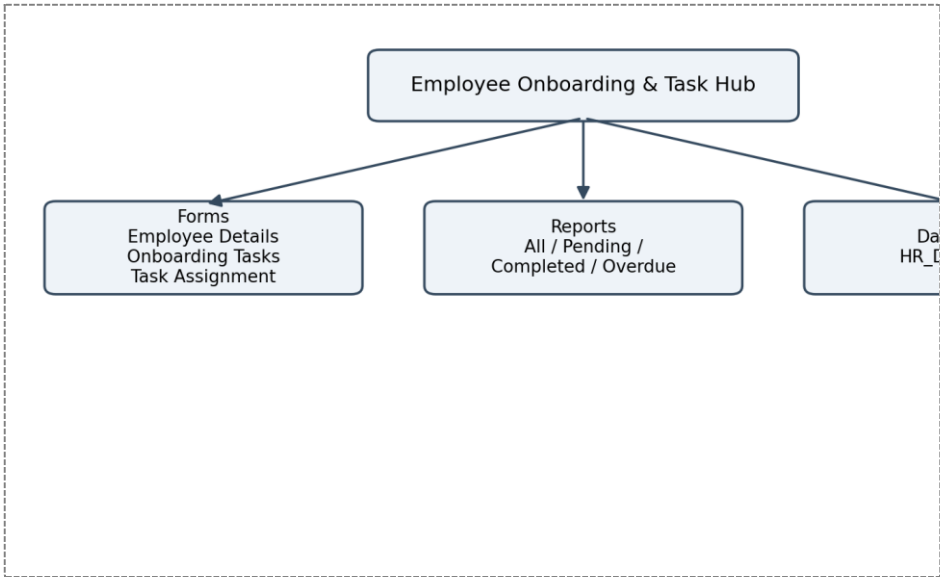


Figure 4.4: Application Navigation Structure

4.6 Access Control Design

Table 4.3: Role–Permission Matrix

Component	HR Administrator	Employee	Portal User
Employee_Details form	Configured – verify access	NOT VERIFIED	Portal access configured; form permissions not verified
Onboarding_Tasks form	Configured – verify access	NOT VERIFIED	Portal access configured; form permissions not verified
Task_Assignment form	Configured – verify access	[NOT VERIFIED]	Portal access configured; field permissions not verified
Reports and HR_Dashboard	Configured – verify access	NOT VERIFIED	NOT VERIFIED

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator using the low-code application builder. The implementation is organised around three forms, six list reports, one HR_Dashboard page and three workflow automations. The dashboard was generated with assistance from Zia AI. The project uses Creator-native email actions and a Deluge statement for completion-date population.

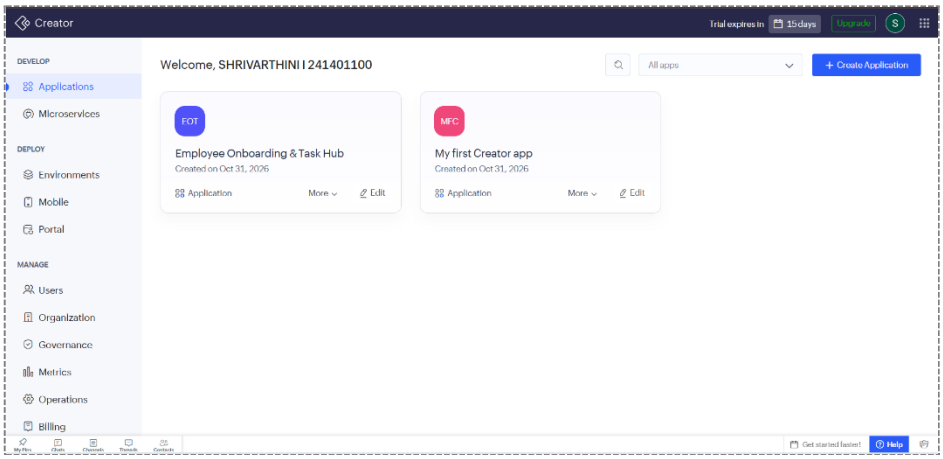


Figure 5.1: Application Home and Navigation Sections (Insert Live Screenshot)

5.2 Forms and Data Capture

Employee_Details stores Employee Name, Email, Department and Employee Status. Onboarding_Tasks stores Task Name and Task Description; any additional fields must be confirmed in the live form. Task_Assignment stores Employee_Details and Onboarding_Tasks lookup values, Assigned_Date, Due_Date, Completion_Date, Task_Status and Remarks. Task_Status uses Not Started, In Progress, Completed and On Hold. Zoho Creator forms are the primary data-entry components of an application [1].

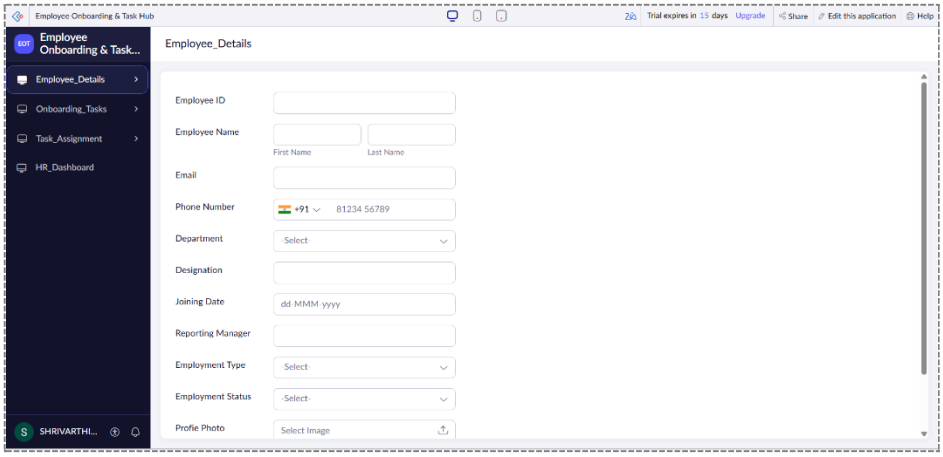


Figure 5.2: Employee_Details Form in the Form Builder (Insert Live Screenshot)

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All_Employees	List	Employee_Details	All employee records	Not reported	Access permissions to verify
All_Onboarding_Tasks	List	Onboarding_Tasks	All onboarding task records	Not reported	Access permissions to verify
All_Task_Assignments	List	Task_Assignment	All assignment records	Not reported	Access permissions to verify
Pending_Tasks	List	Task_Assignment	Pending / incomplete task view; exact criteria to verify	Not reported	Access permissions to verify
Completed_Tasks	List	Task_Assignment	Completed task view	Not reported	Access permissions to verify
Overdue_Tasks	List	Task_Assignment	Overdue view; filter requires final verification	Not reported	Access permissions to verify

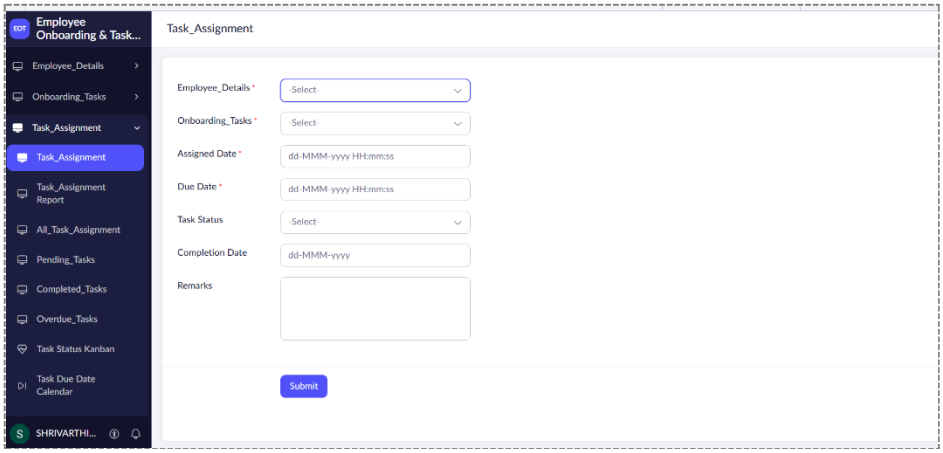


Figure 5.4: All_Task_Assignments Report (Insert Live Screenshot)

5.4 Pages and Dashboards

The HR_Dashboard was generated with assistance from Zoho Creator Zia AI. Its initial layout contains four metrics—Total Employees, Total Onboarding Tasks, Total Task Assignments and Completed Tasks—along with a Task Status Overview doughnut chart, a Department-wise Employee Distribution bar chart, and embedded Pending Tasks and Completed Tasks reports. Planned additions include an Employee Onboarding Status chart, Task Completion Rate KPI and Overdue Onboarding Tasks report; these are not described as implemented. Creator pages can combine components, and reports can be embedded in a page [5], [6].

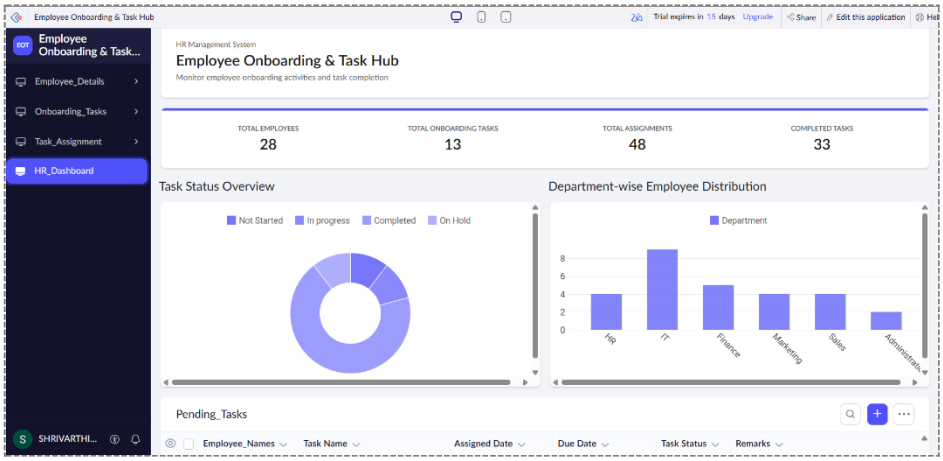


Figure 5.5: HR_Dashboard (Insert Live Screenshot)

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Task_Assignment → workflow when record is edited; condition Task_Status = Completed	Set Completion_Date to current date.	input field assignment; zoho.currentdate
DS-02	Employee_Details → on successful submission workflow	Send welcome email to employee; employee name inserted through Insert Fields.	Creator email action; form field insertion
DS-03	Task_Assignment → schedule based on Due_Date	Send reminder email when due date is reached and Task_Status is not Completed.	Date-based schedule; lookup value; email action

Code 5.1: Deluge Script – Automatic Completion Date

```
// DS-01 | Form: Task_Assignment | Workflow: Auto_Set_Completion_Date
// Trigger: Existing record edited; condition Task_Status = Completed
// Purpose: Populate Completion_Date with the current date.
input.Completion_Date = zoho.currentdate;
```

Code 5.2: Welcome Email Workflow Action (Configured in Creator)

```
// DS-02 | Employee_Details | On successful submission
// Action: Send welcome email using Zoho Creator workflow email
action.
// Employee name is inserted using the Insert Fields option in the
workflow editor.
// No Deluge source text was supplied for this email action.
```

The completion-date workflow runs when an existing Task_Assignment record is edited and the Task_Status condition is Completed. The Deluge statement assigns zoho.currentdate to Completion_Date. Testing confirmed that the date was populated automatically. The other two workflows use Creator email actions: Send_Welcome_Email sends a welcome message after a new employee record is submitted; Due_Date_Reminder sends a reminder on the due date when the assignment is not Completed. The Deluge input keyword provides access to form fields [3], and zoho.currentdate supplies the current date [4].

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Send_Welcome_Email	Form workflow	New Employee_Details record successfully submitted	On successful submission	Send welcome email to employee; name inserted using Insert Fields	DS-02
Auto_Set_Completion_Date	Form workflow	Existing Task_Assignment record edited and submitted	Task_Status = Completed	Set Completion_Date = zoho.currentdate	DS-01
Due_Date_Reminder	Schedule workflow	Based on Due_Date; execute on due date once [7]	Task_Status is not Completed	Retrieve employee email through lookup and send reminder	DS-03

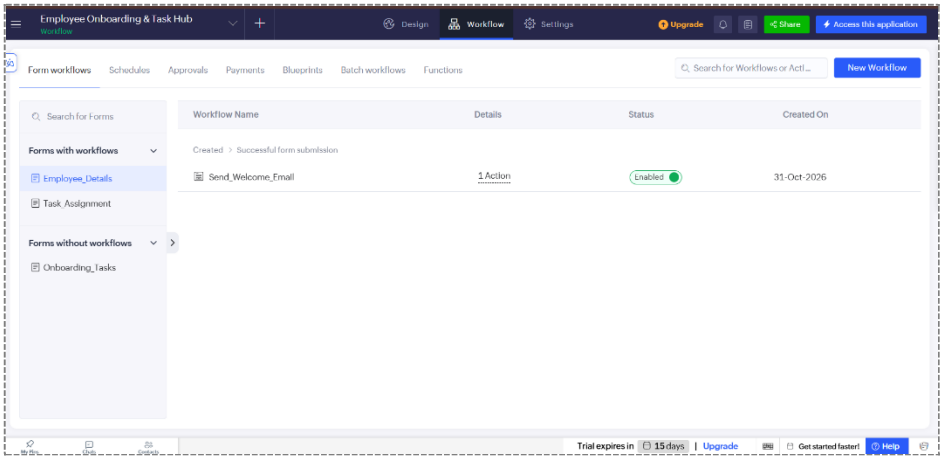


Figure 5.6: Workflow Configuration (Insert Live Screenshot)

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Creator email notification	Zoho Creator email functionality	Workflow email action	Creator account configuration; exact settings not recorded	Welcome email and due-date reminder to employee
External API	Not implemented / not reported	N/A	N/A	No external REST API, CRM sync or payment integration is claimed

Code 5.3: External API Integration – Not Implemented

```
// No external API integration is implemented or reported.
// The application uses native Zoho Creator email workflow actions.
```

5.8 User Management and Role-Based Access

The application is intended primarily for HR administration. The exact user accounts, profiles, role hierarchy and record-level permission settings were not included in the project evidence. Therefore, this report documents the operational HR use case but does not claim that a complete role-based access-control matrix has been configured. Access settings should be verified in the live application before submission.

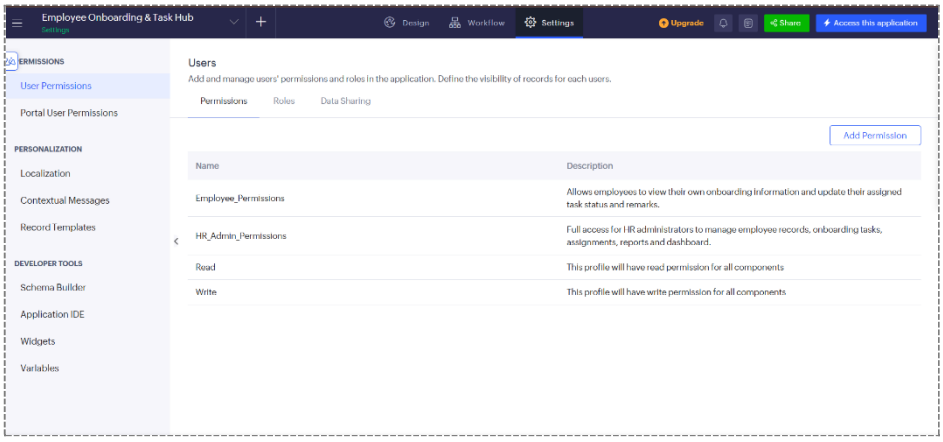


Figure 5.8: Permission Configuration (Insert Screenshot if Configured)

5.9 AI and Analytics Features

Zoho Creator Zia AI was used to generate the HR_Dashboard. The confirmed dashboard contains four summary metrics, two charts and two embedded reports. No predictive model, OCR, sentiment analysis, employee scoring or automated hiring decision feature is claimed.

5.10 Pro-Code Extension

No Zoho Catalyst service, custom JavaScript widget, external REST API or separate pro-code extension is recorded for this project. The application was implemented using Creator forms, reports, pages, workflows and Deluge. This section is retained to match the prescribed report structure; a pro-code extension was not required for the verified scope.

5.11 Mobile and Portal Access

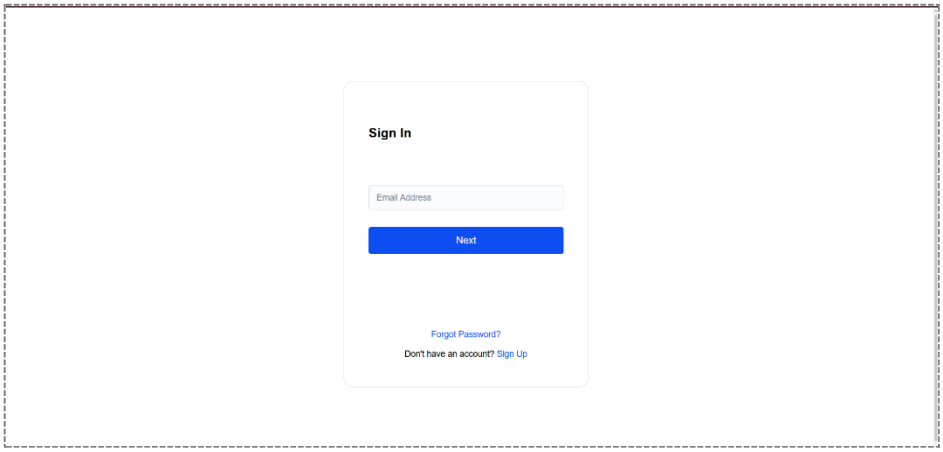


Figure 5.10: Not Applicable – Mobile / Portal Access Not Verified

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing focused on form submission, lookup-based assignment creation, report availability, dashboard display and workflow behaviour. The recorded tests included employee creation, welcome email delivery and name insertion, assignment creation, automatic completion-date population and due-date reminder delivery. The Overdue_Tasks report was created, but its filter requires final verification. Formal load, penetration, access-control and broad negative testing are not claimed.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Create employee record	Submit Employee_Details with available employee information.	Employee record is saved.	Employee record created during testing.	Pass
TC-02	FR-07	Welcome email trigger	Submit a new Employee_Details record.	Welcome email is sent.	Email received.	Pass
TC-03	FR-07	Employee name in welcome email	Inspect received welcome email.	Correct employee name is inserted.	Employee name verified.	Pass
TC-04	FR-02	Create onboarding task	Submit Onboarding_Tasks with known task details.	Task record is saved.	Task records exist in sample data.	Pass
TC-05	FR-03	Create task assignment	Select Employee_Details and Onboarding_Tasks lookup values and submit.	Assignment links to selected records.	Assignment creation tested.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-06	FR-04	Update assignment status	Edit assignment and set Task_Status to Completed.	Status is updated.	Completed status recorded.	Pass
TC-07	FR-08	Automatic completion date	Submit edited assignment with status Completed.	Completion_Date is populated with current date.	Completion date populated automatically.	Pass
TC-08	FR-09	Due-date reminder	Use scheduled assignment due-date workflow with status not Completed.	Reminder email is sent on due date.	Reminder email received in test.	Pass
TC-09	FR-05	All employee report	Open All_Employees.	Employee list is displayed.	Report created and available.	Pass
TC-10	FR-05	All task report	Open All_Onboarding_Tasks.	Task list is displayed.	Report created and available.	Pass
TC-11	FR-05	All assignments report	Open All_Task_Assignments.	Assignment list is displayed.	Report created and available.	Pass
TC-12	FR-06	Pending tasks report	Open Pending_Tasks.	Incomplete tasks are displayed.	Report created and tested.	Pass
TC-13	FR-06	Completed tasks report	Open Completed_Tasks.	Completed tasks are displayed.	Report created and tested.	Pass
TC-14	FR-06	Overdue tasks report	Review Overdue_Tasks filter and compare against due dates/status.	Only overdue incomplete assignments appear.	Report created; filter still requires final verification.	Pending verification

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-15	FR-05	Dashboard metrics and charts	Open HR_Dashboard.	Metrics, charts and embedded reports render.	Dashboard displayed with recorded metrics and components.	Pass
TC-16	FR-01	Negative test: submit employee form with a missing required value	Leave a mandatory field blank and submit.	Creator should prevent invalid submission and display validation feedback.	Not executed / evidence not supplied.	Pending
TC-17	FR-01	Negative test: enter an invalid email address	Enter an invalid email format and submit.	Email validation should reject the invalid value if configured.	Not executed / validation configuration to verify.	Pending
TC-18	NFR-06	Unauthorised access check	Attempt to access a record/report using a user without permission.	Access should be denied according to configured permissions.	Role and permission configuration not verified.	Pending

6.3 Debugging Deluge Scripts

The available project record confirms successful end-to-end tests for the three workflows but does not document a specific Deluge defect, debug log or root cause. The workflow configuration and received email results were used as practical verification evidence. Add an actual workflow log or a real defect and its correction if available; do not invent a debugging incident.

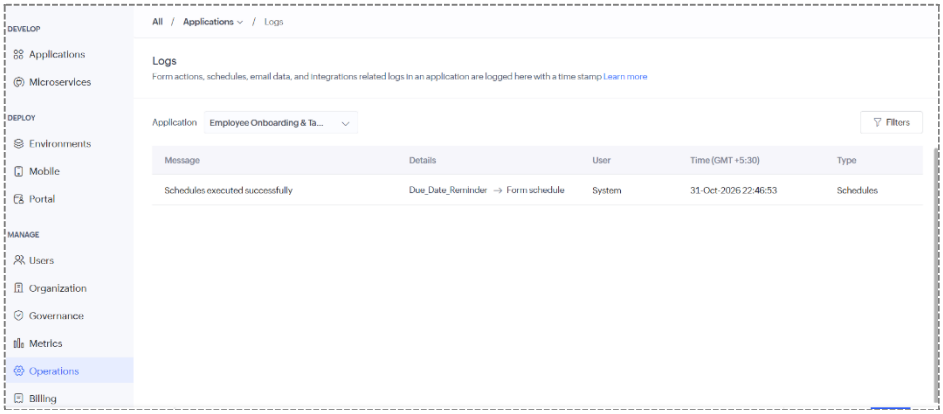


Figure 6.1: Workflow Execution Evidence (Insert Actual Log Screenshot if Available)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	No specific defect record was provided in the project notes.	Not documented.	No fix is claimed. Add a real defect entry if one was encountered.	Not recorded
BUG-02	Overdue_Tasks filter requires final verification.	Filter criteria not yet confirmed.	Review the report criteria against Due_Date and Task_Status.	Open

6.5 User Acceptance Testing

Formal user-acceptance feedback from two independent end users was not included in the project evidence. Table 6.3 is therefore left as a clearly marked record sheet for the actual users to complete after trying the application; feedback has not been fabricated.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	HR Administrator	The application makes it easy to add employee details and assign onboarding tasks. The reports help identify pending and completed tasks.	Improved the organisation of task reports and reviewed the dashboard layout.
User 2	Employee	The application is easy to use. Viewing assigned tasks and updating task status is simple. Due-date reminders are useful.	Reviewed the task assignment interface and clarified task status information.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The project evidence confirms development and testing within Zoho Creator, but does not identify separate Development, Stage and Production environments, a production publishing event, version notes or a rollback exercise. These deployment controls are therefore recorded as items requiring verification rather than completed activities.

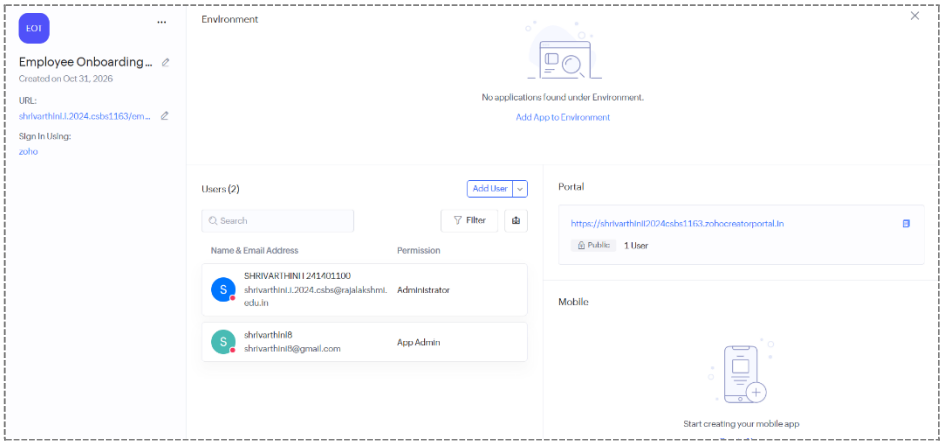


Figure 7.1: Deployment Environment Evidence (Insert if Configured)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	Core forms and lookup relationships configured	Yes
2	Six list reports created	Yes; Overdue_Tasks filter needs verification
3	HR_Dashboard generated and displayed	Yes
4	Welcome email workflow tested	Yes
5	Completion-date workflow tested	Yes
6	Due-date reminder workflow tested	Yes
7	Separate Development / Stage / Production environments verified	Not confirmed
8	Production publishing and rollback plan documented	Not confirmed

7.2 Performance Optimisation

No before-and-after response-time measurements, load-test results or quantified performance benchmarks were recorded. The use of lookup fields reduces

repeated entry of employee and task details, while dedicated list reports provide focused views. These are functional design benefits; no numerical performance gain is claimed.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
No measured optimisation benchmark recorded.	Application-wide	Not measured	Not measured	Performance improvement cannot be quantified.
Centralised lookup-based data capture	Task_Assignment	Employee/task details could be re-entered manually.	Select existing Employee_Details and Onboarding_Tasks records.	Reduces repeated entry and links assignments to master records.
Status-oriented list reports	Pending_Tasks / Completed_Tasks	Manual review of all assignment rows.	Dedicated status reports.	Provides focused operational views; load-time improvement not measured.

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho Creator account access	Platform sign-in; exact account policies to verify.	Unauthorised application access.
Form/report permission profiles	Not evidenced in project notes; verify in live app.	Unauthorised record viewing or modification.
Record-level and field-level restrictions	Not evidenced; not claimed as implemented.	Access to records or fields beyond user responsibility.
Email recipient verification	Welcome and reminder workflows.	Incorrect notification recipient; test addresses should be validated.

The application contains employee-related information, so access permissions must be reviewed before real organisational use. The available project notes do not confirm role profiles, record-level criteria, field-level hiding or encrypted fields. Table 7.3 distinguishes the verified platform access context from controls that still require configuration evidence.

7.4 Governance and Administration

The project evidence does not document an audit-trail review, backup/export procedure, licence-management process, named application administrator or formal change-control record. These governance activities should be completed and recorded before production use. The application uses consistent form, report and workflow names to make the configuration easier to maintain.

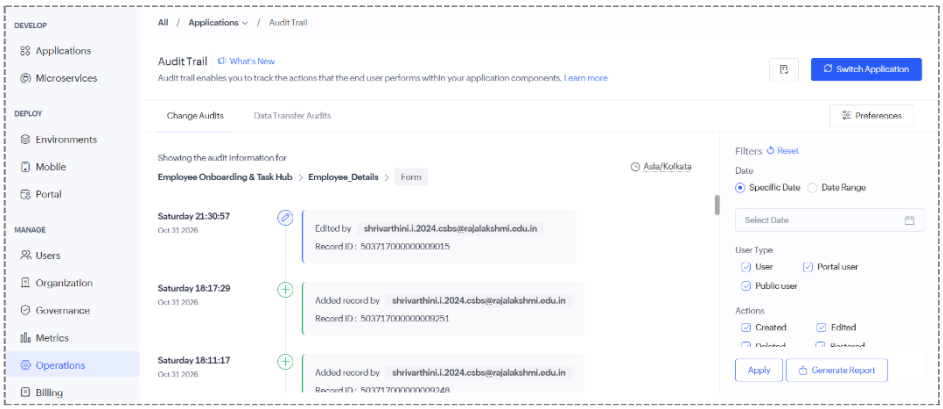


Figure 7.2: Audit Trail / Administration Settings (Insert if Verified)

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A representative onboarding scenario begins with HR entering a new employee in Employee_Details. The welcome-email workflow sends a message to the employee. HR then selects an onboarding task from Onboarding_Tasks and creates a Task_Assignment record using the employee and task lookup fields. The assignment is monitored through the list reports. When the task is marked Completed, the completion-date workflow populates Completion_Date. For an incomplete task reaching its due date, the scheduled reminder workflow sends an email.



Figure 8.1: Employee Record Creation (Insert Live Screenshot)

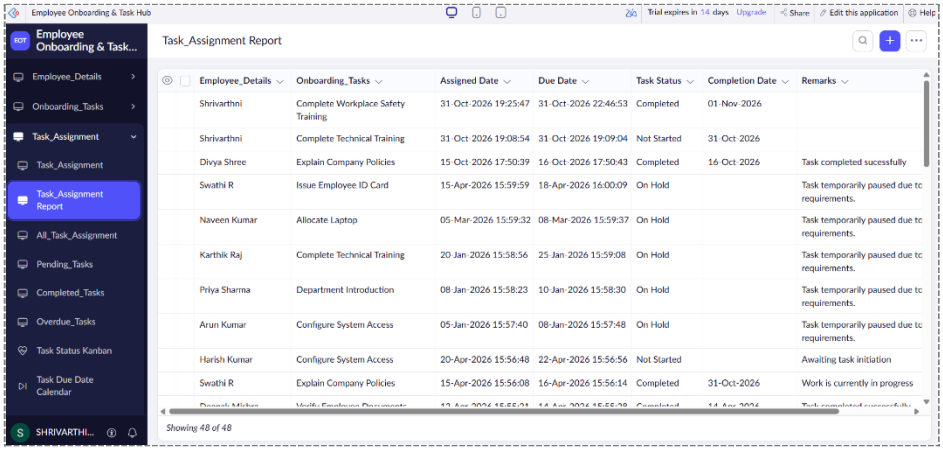


Figure 8.2: Task Assignment and Lookup Selection (Insert Live Screenshot)

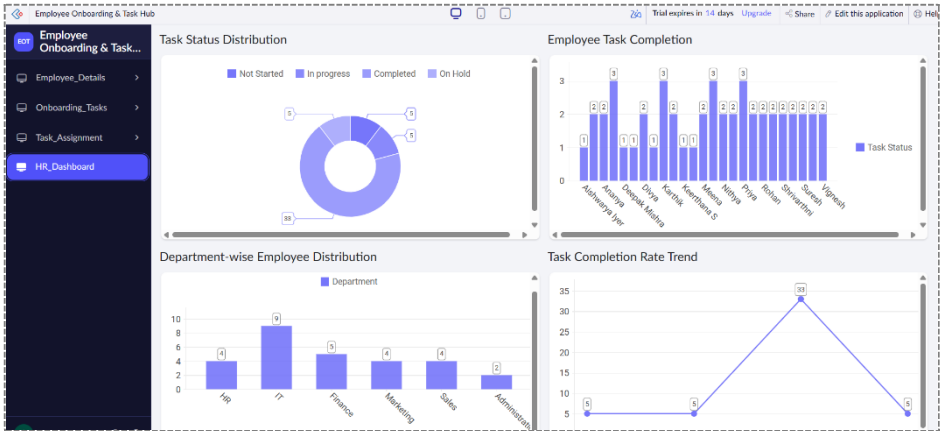


Figure 8.3: Task Completion and Dashboard View (Insert Live Screenshot)

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Maintain employee records	Employee_Details form; All_Employees report	Employee form and report created and tested.
Maintain reusable onboarding tasks	Onboarding_Tasks form; All_Onboarding_Tasks report	Task form and report created.
Link employee and task assignments	Task_Assignment lookup fields	Assignment creation tested.
Automate notifications and completion date	Three workflows; Deluge script DS-01	Welcome email, due-date reminder and completion-date update tested.
Monitor onboarding progress	Six list reports and HR_Dashboard	Dashboard metrics, charts and embedded reports displayed.

The objectives are mapped to the forms, reports, dashboard and workflows in Table 8.1. The principal data-entry, assignment, notification and completion-date objectives were demonstrated in the recorded tests. The overdue report filter remains subject to final verification.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Employee data	May be stored in separate sheets/files.	Employee_Details form.	Centralised employee record store.
Task catalogue	May be maintained separately.	Onboarding_Tasks form.	Reusable task definitions.

Parameter	Existing System	Proposed Application	Improvement
Task allocation	Manual assignment and follow-up.	Task_Assignment with two lookups.	Explicit employee-task relationships.
Progress monitoring	Manual consolidation.	Pending, Completed and All Task Assignment reports.	Dedicated views; actual time savings not measured.

The proposed application replaces disconnected employee, task and assignment tracking with related Creator forms, lookup relationships and dedicated reports. It also adds automatic email notifications and completion-date population. Actual time saved or error-rate reduction was not measured, so the comparison is qualitative.

8.4 Limitations of the Low-Code Approach

The current implementation is a focused onboarding tracker rather than a complete HRMS. The exact field types and additional fields in Onboarding_Tasks require confirmation from the live form. The Overdue_Tasks filter requires final verification. Role permissions, production deployment, external integrations, employee portal access and mobile-specific settings are not evidenced. The current workflow does not document how Completion_Date is handled if a completed task is later changed to another status.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is configured through Zoho Creator's low-code interface: forms, lookup fields, list reports, dashboard components and workflow triggers. Deluge is used for the completion-date assignment. A separate pro-code service was not needed for the confirmed requirements. A future version could consider an external service only if requirements exceed Creator's native capabilities.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Employee Onboarding & Task Hub demonstrates a focused employee onboarding application developed using Zoho Creator. It centralises employee details, onboarding task definitions and task assignments. Lookup fields connect each assignment to the relevant employee and task. Six list reports and an HR_Dashboard provide views of the stored records and task progress. Three workflows automate welcome emails, completion-date population and due-date reminders. Recorded tests confirmed the principal form, notification, date-update and dashboard functions. The Overdue_Tasks filter and several governance and deployment settings still require final verification.

9.2 Future Enhancements

- Verify and refine the Overdue_Tasks report filter and add the planned Overdue Onboarding Tasks dashboard report.
- Add the planned Employee Onboarding Status chart and Task Completion Rate KPI after confirming the required status definitions.
- Configure and test role-based permissions, employee self-service access and a documented production deployment process.

9.3 Learning Outcomes and Career Relevance

The project provided practical experience in low-code application development, form design, lookup-based data modelling, report configuration, dashboard composition, workflow scheduling and Deluge scripting. These skills are relevant to low-code development, business process automation and application support roles. The project also reinforced the importance of testing each workflow and documenting only features verified in the live application.

REFERENCES

[1] Zoho Corporation, “Understanding forms,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/forms/the-concept-of-forms/articles/form-component. [Accessed: 01-Nov-2026].

[2] Zoho Corporation, “Understand Lookup Field,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/forms/add-and-manage-fields/articles/fields-lookup-understand. [Accessed: 01-Nov-2026].

[3] Zoho Corporation, “Accessing Form Fields using input,” Deluge Help. [Online]. Available: https://www.zoho.com/deluge/help/data-access/accessing-form-fields.html. [Accessed: 01-Nov-2026].

[4] Zoho Corporation, “System Variables,” Zoho Developer Help. [Online]. Available: https://www.zoho.com/developer/help/scripting/system-variables.html. [Accessed: 01-Nov-2026].

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[6] Zoho Corporation, “Understanding reports in a page,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/pages/reports-in-pages/articles/understand-reports-in-a-page. [Accessed: 01-Nov-2026].

[7] Zoho Corporation, “Understanding schedules,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/schedule-workflows/articles/understand-schedules. [Accessed: 01-Nov-2026].

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Yes	Employee Onboarding & Task Hub	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Not verified	Three forms; exact field types not fully recorded	5.2	Tables 4.2–4.4
II	Mandatory field validation	Not verified	Verify in live form configuration	5.2	—
II	Lookup field relationships	Yes	Task_Assignment	4.2–4.3	Fig. 4.2
II	Subform implementation	No / not reported	Not used in verified scope	5.2	—
II	List reports	Yes	Six named list reports	5.3	Table 5.1
II	Advanced report types (Kanban / Calendar / Pivot)	No / not reported	List reports only	5.3	Table 5.1
II	Pages and dashboard	Yes	HR_Dashboard	5.4	Fig. 5.5
II	Charts and KPI panels	Yes	Four metrics and two charts	5.4	Fig. 5.5
II	Embedded reports in page	Yes	Pending Tasks and Completed Tasks	5.4	Fig. 5.5
II	Deluge form scripting	Yes	Task_Assignment completion date	5.5	Code 5.1
II	Form workflows	Yes	Send_Welcome_Email; Auto_Set_Completion_Date	5.6	Table 5.3
II	Schedule workflow	Yes	Due_Date_Reminder	5.6	Table 5.3

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Approval process	No / not reported	Not used	5.6	—
II	Blueprint	No / not reported	Not used	5.6	—
III	Zoho CRM integration	No / not reported	No external CRM integration	5.7	—
III	External REST API integration	No / not reported	No external API integration	5.7	—
III	Connections / OAuth integration	Not verified	No external connection reported	5.7	—
III	User roles and profiles	Not verified	Review live application settings	5.8	—
III	Record-level permissions	Not verified	Review live application settings	5.8	—
III	Field-level permissions	Not verified	Review live application settings	5.8	—
III	Zia AI / AI-assisted feature	Yes	Zia AI-assisted HR_Dashboard generation	5.9	___
III	Predictive AI / OCR / sentiment analysis	No / not reported	Not used	5.9	—
IV	Development / Stage / Production environments	Not verified	Separate environments not documented	7.1	—
IV	Production publishing and rollback	Not verified	Not documented	7.1	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Performance optimisation benchmark	No / not measured	No benchmark recorded	7.2	—
IV	Security controls review	Pending verification	Permission configuration to verify	7.3	—
IV	Audit trail / governance	Not verified	Review Creator administration settings	7.4	—
V	Zoho Catalyst extension	No / not reported	Not used	5.10	—
V	JavaScript widget / Creator API	No / not reported	Not used	5.10	—
V	Mobile app / portal access	Not verified	No mobile or portal configuration evidence	5.11	—
V	User manual and learning reflection	Partly	Appendix D; learning outcomes in 9.3	9.3 / App. D	—

Confirmed as implemented: refer to the "Done" column in Table A.1; features marked Not Verified are not counted as implemented.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Chapters 1, 2 and 4	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Chapters 4, 5 and 6	3	
CO3	Apply integration, user management, AI and analytics features where appropriate.	Sections 5.7–5.9; unsupported features explicitly identified	2	
CO4	Evaluate, test, deploy and govern low-code applications.	Chapters 6, 7 and 8; deployment controls require verification	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Task_Assignment – Auto_Set_Completion_Date

```
// Script ID: DS-01
// Location: Task_Assignment → Auto_Set_Completion_Date workflow
// Condition: Task_Status = Completed
// Purpose: Set the completion date to the current date.
input.Completion Date = zoho.currentdate;
```

C.2 Email Workflow Actions – Send_Welcome_Email and Due_Date_Reminder

```
// Script ID: DS-02 / DS-03
// Location: Employee_Details welcome email and Task_Assignment due-
date reminder workflows
// These are configured workflow email actions. No separate Deluge
source code was supplied.
// Verify the live workflow configuration before submission.
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Access the application through the Zoho Creator web interface using the authorised Zoho account. Open the Employee Onboarding & Task Hub application from the Creator workspace. Mobile and portal access are not confirmed in this project and should not be assumed.

D.2 Instructions for HR Administrator / Application User

- Open Employee_Details and add a new employee using the configured fields. Confirm that the welcome email is received.
- Open Onboarding_Tasks to maintain reusable task definitions.

D.3 Employee Email Notifications

- Open Task_Assignment, select the employee and task lookup values, enter the dates and status, and save.
- Employees receive a welcome email after a new Employee_Details record is submitted.

Employees may receive a due-date reminder for an assignment that has reached its due date and is not marked Completed.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SHRIVARTHINI I	Register No.	241401100
Project Title	Employee Onboarding & Task Hub	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SHRIVARTHINI I	Register No.	241401100
Department / Section	Computer Science and Business Systems	Date of Viva	
Application URL Verified	[YES / NO]	Demo Duration	[_____ min]

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____